Supplementary File S3: Paired Entry-Path Integration Check

This file reports a deterministic simulator check of two software entry routes. It validates operation dispatch and device-write evidence; it does not evaluate agent reasoning, multi-agent coordination, or process improvement. The accompanying evidence bundle contains the runner record, fixture-trace summary and hashes, and local service logs.

1 Setup and Procedure

The test used one isolated loopback-only application and simulator instance. One deterministic OMP-RPC fixture supplied a predeclared `recipe_trial` call through the registered task route; a second call used the direct REST operation endpoint. No LLM inference, external model request, or physical PLC was involved. Each route used its own line, product, recipe, control and DAQ nodes, channel, agent record, and synthetic administrator approval. Within each scene, the two routes shared a normalized profile of protocol, signal mapping, engineering range, step limit, target band, and admissible candidate set; they used different candidates from that set. Consequently, this is a paired scene-profile check, not an identical-entity or matched process trial.

Before each write, the runner read the control value through a separate DCW read operation. It then recorded the approval, recipe-trial operation, and platform acknowledgement, followed by another DCW read. The runner also checked that the trial and acknowledgement records linked to the same operation run, that the acknowledgement was verified, and that simulator and application processes and loopback ports were released after execution.

2 Results

Table 1: Pre-write value, requested setpoint, and separate post-write device readback for each route.

Scene and protocol	Entry route	Transition	Platform receipt and readback
Cast-film, OPC UA	Registered task	120→125 rpm	1/1 verified; readback 125 rpm
Cast-film, OPC UA	Direct REST	120→130 rpm	1/1 verified; readback 130 rpm
Annealing, Modbus TCP	Registered task	730→735°C	1/1 verified; readback 735°C
Annealing, Modbus TCP	Direct REST	730→740°C	1/1 verified; readback 740°C

All four route operations changed the observed control value to the requested candidate. All four platform acknowledgements reported one verified write out of one, and each separate post-write read matched the setpoint. The complete runner reported 24 passing checks, including loopback isolation, fixture selection, platform receipt linkage, device readback, process cleanup, and port release.

3 Interpretation and Limits

The check supports the claim that the registered task entry and direct REST entry can each reach a bounded, approved recipe-trial path and produce a verified simulator write under the tested scene profiles. The route records used separately instantiated production entities, and their candidate values differed. The registered fixture trace contained one tool call and one result per scene; the associated task state remained `SUBMITTED`. The check did not test task completion, process response, rejected candidates, or agent decisions. It therefore establishes neither general route equivalence, multi-agent capability, process benefit, safety certification, nor physical-plant performance. A same-entity attempt verified its registered route but timed out in direct REST; it is excluded and retained in the evidence bundle. The execution-time runner source was not preserved, limiting exact reproduction to the recorded provenance hashes.